

2.0 Literature Review

2.1 Optimizer

Zhou et al. (2024) examines the optimization behaviour of the AdamW optimizer, with a focus on its convergence characteristics and generalization performance. The study provides theoretical and empirical insights into the advantages of AdamW, particularly when combined with effective learning rate scheduling strategies such as cosine decay and linear warmup, compared with conventional optimization approaches based on Adam and step-decay schedules.

AdamW and Cosine Decay Learning Rate

- **Decoupled Weight Decay:** AdamW introduces a decoupled weight decay, which enhances convergence and generalization compared to standard Adam and l2-Adam.
- **Performance with Learning Rate Schedules:** The paper suggests that AdamW, when paired with cosine decay learning rates, can yield superior results in terms of convergence speed and final accuracy.

Comparison with Standard Adam

- **Convergence Analysis:** AdamW is shown to converge more effectively than standard Adam, especially in over-parameterized networks.
- **Generalization Error:** AdamW exhibits smaller generalization errors, which is critical for tasks requiring high accuracy.

While the advantages of AdamW are significant, some studies suggest that traditional methods like SGD with step-decay can still outperform adaptive methods in certain scenarios, particularly in terms of final convergence quality (Lin et al., 2024).

2.2 Augmentations

Hendrycks and Dietterich (2019) investigated the robustness of deep neural networks under realistic image variations and introduced the ImageNet-C and ImageNet-P benchmarks to evaluate model performance beyond conventional clean-image accuracy. ImageNet-C measures the degradation of model performance under common image corruptions, such as noise, blur, weather effects, and digital distortions, while ImageNet-P evaluates prediction stability under small natural perturbations.

The authors demonstrated that achieving high accuracy on clean datasets does not necessarily indicate strong robustness in real-world conditions. Modern neural networks can achieve excellent performance when evaluated on standard test images but experience significant performance degradation when exposed to corrupted or altered inputs. These findings highlight that robustness is an important property separate from classification accuracy and emphasize the need for developing models that can learn more stable and generalizable representations for real-world applications.

Following this direction, Hendrycks et al. (2020) proposed **AugMix**, a data augmentation technique designed to improve model robustness against common corruptions without modifying the underlying network architecture. Unlike conventional augmentation methods that typically apply individual transformations independently, AugMix generates training samples by combining multiple augmentation operations through randomly selected transformation chains. This process exposes the model to a wider range of possible image variations and encourages learning of more invariant features.

The authors evaluated AugMix using corruption robustness benchmarks and demonstrated that models trained with AugMix achieved improved performance on corrupted images while maintaining comparable accuracy on clean datasets. In addition, AugMix improved model calibration by producing more reliable confidence estimates when encountering uncertain or degraded inputs. These results indicate that input-level processing strategies can effectively enhance model

reliability without requiring architectural modifications or additional computational complexity.

The studies by Hendrycks and Dietterich (2019) and Hendrycks et al. (2020) emphasize the importance of addressing robustness at the data and preprocessing levels. While architectural improvements remain an important research direction, these works demonstrate that the quality and diversity of input representations significantly influence model generalization. For image recognition systems operating under real-world conditions, preprocessing and augmentation techniques provide practical approaches to reduce the impact of input degradation.

2.3 Knowledge distillation

Deep learning models have achieved significant improvements in recognition tasks by increasing model size and using ensemble strategies. However, deploying these large models in practical applications remains challenging due to their high computational and memory requirements. Hinton et al. (2015) proposed knowledge distillation as a method to compress the knowledge of a complex model into a smaller network while maintaining competitive performance. Instead of training the student model only using the original hard labels, the proposed method trains the student using the soft probability outputs generated by the teacher model. These soft targets contain additional information about the similarity between different classes, allowing the student model to learn richer representations beyond the ground-truth labels.

The key concept of knowledge distillation is that a trained teacher network can provide guidance to a smaller student network by transferring its learned knowledge. The teacher model produces probability distributions over possible classes using a temperature scaling mechanism, where higher temperatures generate softer probability distributions. The student model then learns to reproduce these probability patterns, enabling it to achieve improved generalization compared with conventional training using only hard labels.

The authors demonstrated that knowledge distillation could effectively reduce model complexity while preserving model performance. Experiments on image

classification and speech recognition tasks showed that compact student models trained through distillation could achieve comparable or improved accuracy compared with independently trained models. The study also introduced specialist models, where smaller networks were trained to focus on difficult fine-grained classes that were frequently confused by the main model, further improving prediction performance.

The contribution of this work is significant because it established a practical approach for deploying deep learning models in environments with limited computational resources. Rather than relying solely on increasing model size, knowledge distillation provides an alternative strategy by transferring learned representations from larger models into efficient architectures. This approach has since become widely adopted in applications requiring lightweight models, including mobile vision systems, embedded devices, and edge artificial intelligence applications.

For resource-constrained computer vision applications, knowledge distillation provides an important foundation for balancing accuracy and computational efficiency. This concept is particularly relevant for deploying deep learning models on embedded platforms, where memory usage, processing power, and inference speed are limited. Although knowledge distillation focuses on model compression rather than input improvement, it demonstrates that optimization strategies beyond architectural modification can improve the practicality of deep learning systems.

2.4 Regularization

2.4.1 MixUp

Zhang et al. (2017) proposed mixup, a data augmentation technique designed to improve the generalization and robustness of deep neural networks. The authors argued that conventional training based on empirical risk minimization (ERM) often causes models to memorize training data, resulting in overfitting and poor generalization to unseen samples. To address these limitations, mixup generates virtual training examples by linearly interpolating pairs of input images and their corresponding labels. This encourages the model to learn smoother decision

boundaries and capture more generalized relationships between different classes rather than memorizing individual training samples.

Unlike traditional data augmentation methods that apply geometric or photometric transformations to a single image, mixup combines two different training samples to produce a new training example with a weighted combination of their labels. This simple interpolation strategy acts as a regularization technique by exposing the model to a broader distribution of training data. As a result, the network learns more stable feature representations and becomes less sensitive to small variations in the input data.

The effectiveness of mixup was evaluated on several benchmark datasets, including ImageNet, CIFAR-10, CIFAR-100, Google Commands, and selected UCI datasets. Experimental results demonstrated that models trained using mixup consistently achieved improved classification accuracy and better generalization performance compared with conventional training methods. Furthermore, the study showed that mixup enhanced robustness against noisy labels and adversarial examples while reducing the tendency of neural networks to memorize corrupted training data. These findings indicate that a simple data augmentation strategy can significantly improve model robustness without increasing model complexity or requiring modifications to the network architecture.

The contribution of Zhang et al. (2017) established mixup as a simple yet effective regularization technique for deep learning. By encouraging linear behavior between training examples, the method improves generalization and robustness while maintaining ease of implementation. The success of mixup has also inspired numerous subsequent data augmentation techniques, highlighting its lasting impact on the development of robust deep learning models.

2.4.2 Exponential Moving Average (EMA)

Tarvainen and Valpola (2017) proposed Mean Teacher, a semi-supervised learning approach that improves model performance by effectively utilizing unlabeled data during training. The authors addressed the challenge that deep neural networks typically require large amounts of labeled data to achieve high performance, while

obtaining labeled datasets can be expensive and time-consuming. To overcome this limitation, semi-supervised learning methods aim to exploit both labeled and unlabeled data by encouraging models to learn consistent representations from different variations of the same input.

The Mean Teacher framework is based on the concept of consistency regularization, where a model is trained to produce similar predictions when the input data is subjected to different perturbations. The method consists of two networks: a student model, which is updated using standard gradient-based optimization, and a teacher model, which provides more stable prediction targets for the student. Unlike previous approaches such as Temporal Ensembling, which store and average predictions for individual training samples, Mean Teacher applies an exponential moving average (EMA) to the model weights. This allows the teacher model to represent an averaged version of the student network throughout training, resulting in more stable and reliable predictions.

The main advantage of Mean Teacher is that the teacher network provides improved learning targets by reducing the uncertainty and instability of individual model predictions. By minimizing the difference between student and teacher outputs for unlabeled samples, the network learns representations that remain consistent under different input perturbations. This process improves generalization and reduces the dependency on large amounts of labeled data. Furthermore, the weight-averaging strategy avoids the computational limitations of maintaining prediction histories for all training samples, making the approach more scalable for larger datasets.

The authors evaluated Mean Teacher on several benchmark datasets, including SVHN, CIFAR-10, and ImageNet, and demonstrated that the proposed approach achieved significant improvements compared with previous semi-supervised learning methods. The experimental results showed that Mean Teacher could achieve competitive performance with substantially fewer labeled examples by effectively incorporating unlabeled data during training. These findings demonstrate that consistency-based learning strategies can improve the efficiency of deep learning models when labeled data availability is limited.

The contribution of Tarvainen and Valpola (2017) established Mean Teacher as an important framework in semi-supervised deep learning by demonstrating that weight-averaged teacher models can provide more reliable training targets compared with conventional single-network training. The study highlights the effectiveness of utilizing unlabeled data through consistency regularization and provides a foundation for subsequent research in semi-supervised learning and teacher-student training approaches.

2.5 Quantization

Fan et al. (2020) investigated the challenge of deploying deep neural networks in resource-constrained environments by focusing on extreme model compression through quantization. Although modern deep learning models achieve high accuracy, their large memory requirements and computational costs limit their practical deployment on edge devices. Quantization has become an effective model compression technique by reducing the precision of model parameters from high-precision floating-point representations to lower-bit formats. However, directly applying quantization after training can introduce significant accuracy degradation because the original model is not optimized to operate under reduced numerical precision.

To address this limitation, Quantization-Aware Training (QAT) has been introduced as a method that incorporates quantization effects during the training process. Instead of training a full-precision model and compressing it afterward, QAT simulates quantization during training, allowing the network to adapt to the errors introduced by reduced precision. During QAT, the forward pass uses quantized weights while the gradients are approximated to enable optimization, commonly through the Straight-Through Estimator (STE). This approach allows the model to learn parameters that are more resilient to quantization-induced errors and generally achieves better accuracy compared with conventional Post-Training Quantization (PTQ).

However, Fan et al. (2020) highlighted that conventional QAT approaches face challenges when applied to extreme compression levels. While QAT is effective for

common low-bit quantization such as INT8, more aggressive compression introduces larger approximation errors and makes gradient estimation through the STE increasingly inaccurate. To overcome this limitation, the authors proposed Quantization Noise Training, which extends the concept of QAT by introducing quantization noise into only a randomly selected subset of weights during each training iteration. This strategy allows the remaining weights to maintain accurate gradient updates, resulting in more stable optimization while still encouraging the model to become robust against quantization effects.

The experimental results demonstrated that quantization noise training could achieve extreme compression while maintaining competitive model accuracy. The authors evaluated their approach on both computer vision and natural language processing tasks, showing that highly compressed models could retain performance close to their original full-precision counterparts. These results demonstrate that training strategies that consider quantization effects during optimization are essential for achieving efficient and accurate compressed neural networks.

The work of Fan et al. (2020) emphasizes the importance of adapting neural networks during training rather than treating quantization as a separate post-processing step. By incorporating quantization considerations into the learning process, QAT and its extensions enable deep learning models to achieve improved efficiency while minimizing accuracy loss. This approach has become an important technique for deploying neural networks in applications where memory, computational power, and energy consumption are limited.

References

- Antti Tarvainen, & Harri Valpola. (2017). Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.1703.01780>
- Fan, A., Stock, P., Graham, B., Grave, É., Rémi Gribonval, Hervé Jegou, & Joulin, A. (2021). Training with Quantization Noise for Extreme Model Compression. HAL (Le Centre Pour La Communication Scientifique Directe).
- Hendrycks, D., & Dietterich, T. G. (2019). Benchmarking Neural Network Robustness to Common Corruptions and Perturbations. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.1903.12261>
- Hendrycks, D., Mu, N., Cubuk, E. D., Zoph, B., Gilmer, J., & Lakshminarayanan, B. (2020). AugMix: A Simple Data Processing Method to Improve Robustness and Uncertainty. In arXiv.org. <https://doi.org/10.48550/arXiv.1912.02781>
- Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the Knowledge in a Neural Network. In arXiv.org. <https://doi.org/10.48550/arXiv.1503.02531>
- Lin, Z., Zhang, S., Zhou, Y., Wang, H., & Wang, S. (2024). *Learning rate burst for superior SGDM and AdamW integration*. <https://doi.org/10.3233/jifs-239157>
- Zhang, H., Cisse, M., Dauphin, Y. N., & Lopez-Paz, D. (2017). mixup: Beyond Empirical Risk Minimization. arXiv (Cornell University). <https://doi.org/10.48550/arxiv.1710.09412>
- Zhou, P., Xie, X., Lin, Z., & Yan, S. (2024). Towards Understanding Convergence and Generalization of AdamW. IEEE Transactions on Pattern Analysis and Machine Intelligence, 46(9), 6486–6493. <https://doi.org/10.1109/tpami.2024.3382294>